

Introduction to Node.js, NPM, Modules & File System Operations

What is Node.js?

Node.js is an open-source, cross-platform JavaScript runtime built on the V8 engine. It runs JavaScript outside the browser and is used to build fast, scalable server-side applications.

Features:

- Event-driven
- Non-blocking I/O
- Fast and lightweight
- Cross-platform

Why Learn Node.js?

- JavaScript for frontend and backend
- High performance
- Large package ecosystem
- Suitable for APIs, chat apps, dashboards and e-commerce.

Installing Node.js

Download the LTS version from nodejs.org.

Commands:

```
node -v
```

```
npm -v
```

NPM

NPM (Node Package Manager) is used to install, update, remove and manage project dependencies.

Common commands:

```
npm init -y
```

```
npm install express
```

```
npm uninstall express
```

```
npm update
```

package.json

Stores project information such as name, version, scripts, dependencies, author and license.

Modules

Modules are reusable pieces of code.

Types:

- Core Modules (fs, http, path, os)
- Local Modules
- Third-party Modules

Importing Modules

Example:

```
const fs = require('fs');  
module.exports = add;
```

File System Operations

The fs module is used to create, read, write, append, rename and delete files.

Examples:

```
fs.readFile()  
fs.writeFile()  
fs.appendFile()  
fs.rename()  
fs.unlink()
```

Synchronous vs Asynchronous

Synchronous operations block execution.

Asynchronous operations are non-blocking and recommended for real-world applications.

Summary

Introduction to Node.js, NPM, package.json, Modules, File System operations and synchronous vs asynchronous programming.